الأكاديمية العربية للعلوم والتكنولوجيا والنقل البحري
Arab Academy for Science, Technology & Maritime Transport
College of Computing and Information Technology

CAI3103/CS460

DEEP LEARNING

LECTURE-5

Key Performance Metrics for ML and DL

Performance metrics for Machine Learning and Deep Learning

Performance metrics are crucial in machine learning and deep learning for several reasons

Model Evaluation:

- **Assessing Accuracy:** They help quantify how well a model performs on a given task.
- **Identifying Weaknesses:** They pinpoint areas where the model struggles, enabling targeted improvements.
- **Comparing Models:** They facilitate comparison between different models to select the best-performing one.

Model Improvement:

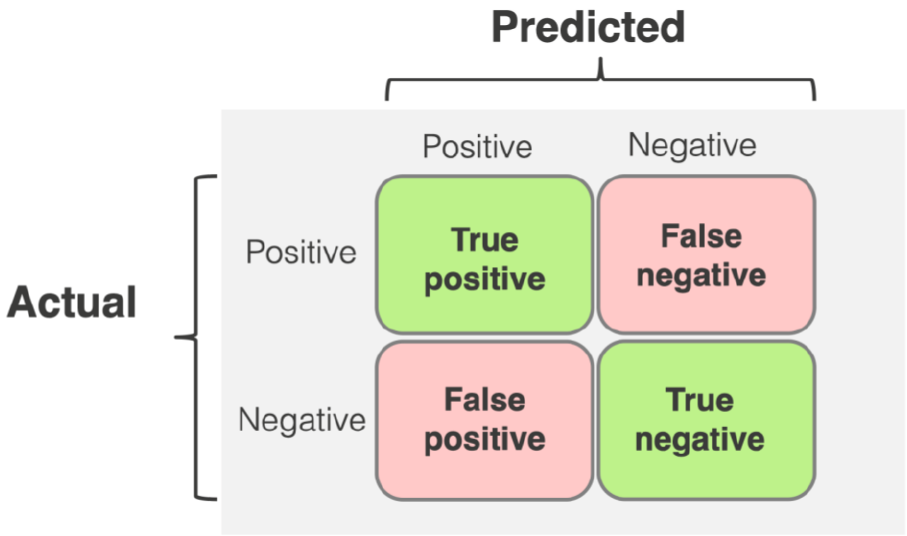
- **Hyperparameter Tuning:** Metrics guide the optimization of hyperparameters to enhance model performance.
- **Feature Engineering:** They help identify relevant features that contribute significantly to the model's accuracy.
- **Algorithm Selection:** They assist in choosing the most suitable algorithm for a specific problem.

Real-world Deployment:

- **Reliability:** Metrics ensure that models deployed in real-world applications meet specific performance thresholds.
- **Risk Assessment:** They help assess the potential risks and consequences of model errors.
- **Decision Making:** They provide quantitative evidence to support decision-making based on model predictions.
- **Benchmarking:** Metrics allow researchers to compare their models to state-of-the-art solutions.

Confusion Matrix

A **confusion matrix** is a visualization method for classifier algorithm results. More specifically, it is a table that breaks down the number of ground truth (actual) instances of a specific class against the number of predicted class instances.



		Predicted Values	
		Positive (1)	Negative (0)
Actual Values	Positive (1)	TP	FN
	Negative (0)	FP	TN

Accuracy

Accuracy is a measure of **how close** a given set of guessing from our model are closed to their true value.

$$\text{Accuracy} = \frac{\#\ \text{Correct classifications}}{\#\ \text{All classifications}}$$

- If a classifier make 10 predictions and 9 of them are correct, the accuracy is 90%.

- Accuracy is a measure of **how well** a binary classifier correctly identifies or excludes a condition.
- It's the **proportion of correct predictions among the total number of cases examined.**

Accuracy

$$accuracy = \frac{TP + TN}{TP + FN + TN + FP}$$

Overall ability of a model to make the correct classification

- TN: True Negative (correctly identified negative cases)
- FP: False Positive (incorrectly identified as positive)
- TP: True Positive (correctly identified positive cases)
- FN: False Negative (incorrectly identified as negative)

Classification case: metrics for skewed classes

Disease dichotomic classification example

Train logistic regression model $h(\boldsymbol{x})$, with $y = 1$ if disease, $y = 0$ otherwise.

Find that you got 1% error on test set (99% correct diagnoses)

Only 0.50% of patients **actually have** disease

The $y = 1$ class has very few examples with respect to the $y = 0$ class

If I use a predictor that **predicts always the 0 class**, I get 99.5% of accuracy!!

For **skewed classes**, the accuracy metric can be deceptive

Precision and recall

Suppose that $y = 1$ in presence of a **rare class** that we want to detect

Precision *(How much we are precise in the detection)*

Of all patients where we predicted $y = 1$, what fraction actually has the disease?

$$\frac{\text{True Positive}}{\text{\# Predicted Positive}} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

Recall *(How much we are good at detecting)*

Of all patients that actually have the disease, what fraction did we correctly detect as having the disease?

$$\frac{\text{True Positive}}{\text{\# Actual Positive}} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

Confusion matrix

Actual class

Predicted class

	1 (p)	0 (n)
1 (Y)	True positive (TP)	False positive (FP)
0 (N)	False negative (FN)	True negative (TN)

Precision and recall

Precision	$= \frac{TP}{TP + FP}$	$\frac{True\ Positive}{True\ Positive + False\ Positive}$	Ability of a classification model to make correct predictions within the positive class
Sensitivity (Recall)	$= \frac{TP}{TP + FN}$	$\frac{True\ Positive}{True\ Positive + False\ Negative}$	Ability to correctly identity positive labels

F-measure (F1-Score)

- The F-score is a single metric that combines precision and recall for calculating the harmonic mean.
- It's particularly useful to balance both metrics, as a high F-score indicates a good balance between precision and recall.

F-score	$\frac{2 \times Precision \times Sensitivity}{Precision + Sensitivity}$	Harmonic mean of sensitivity and precision

Specificity

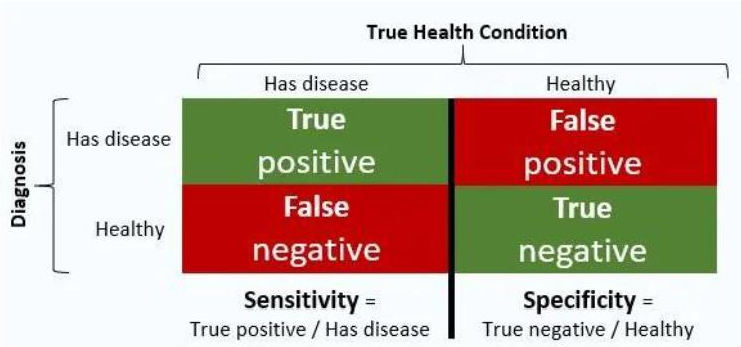
- The Specificity measures the proportion of actual negative cases that are correctly identified as such, in order to identify how well a model can correctly identify negative instances.
- A high specificity value indicates that the model is good at correctly identifying negative cases. (This is crucial in scenarios where false positives are costly, such as medical diagnosis or fraud detection)
- A low specificity value implies that the model is prone to incorrectly classifying negative instances as positive. (This can lead to unnecessary actions or alarms, which can be costly or inconvenient)

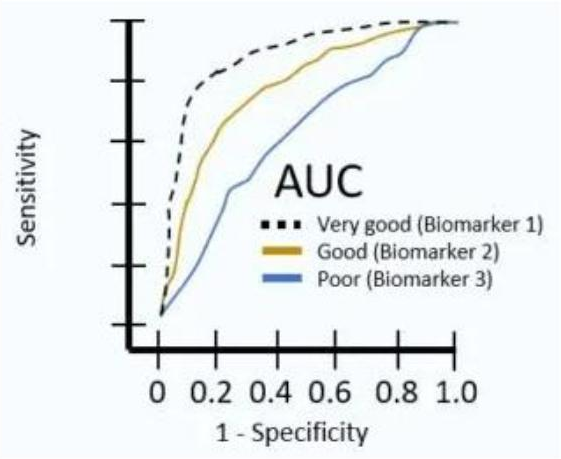
Specificity	$= \frac{TN}{TN + FP}$	$\frac{True\ Negative}{True\ Negative + False\ Positive}$	Ability to correctly identity negative labels

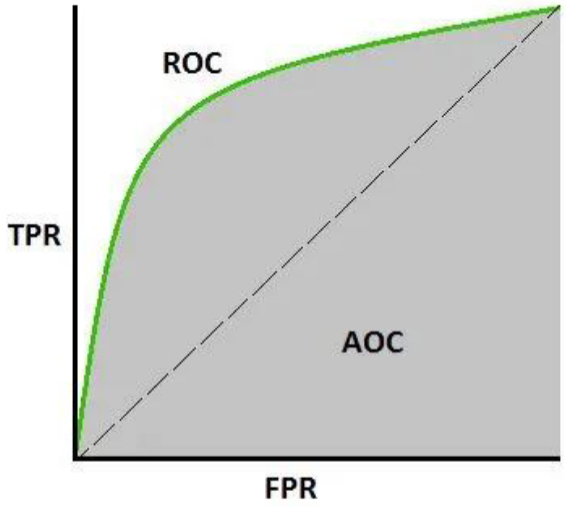
AUC — ROC Curve

- The Receiver Operator Characteristic (ROC) curve is an evaluation metric for binary classification problems. It is a probability curve that plots the TPR (True Positive Rate) or "sensitivity" against FPR (False Positive Rate) or "1-specificity".
- The Area Under the Curve (AUC) is the measure of the ability of a classifier to distinguish between classes and is used as a summary of the ROC curve.

- The ROC curve is plotted with TPR against the FPR where TPR is on the y-axis and FPR is on the x-axis.





AUC — ROC Curve

- AUC-ROC curve is a performance measurement for the classification problems at various threshold settings.
- ROC is a probability curve, and AUC represents the degree or measure of separability. It tells how much model is capable of distinguishing between classes.
- Higher the AUC, better the model is at predicting 0s as 0s and 1s as 1s (distinguishing between patients with the disease and no disease).